

Year 11 Extension 1 - Preliminary Examination 2005

QUESTION 1: START A NEW PAGE

- | | Marks |
|---|--------------|
| (a) Find the acute angle to the nearest degree between the lines with the equations $\sqrt{3}x + y = 2$ and $x - y = 0$. | 3 |
|
 | |
| (b) How many ways can the ten letters of the word REGULATION be arranged in a line if the vowels A, E, I, O, U occur: | |
| (i) All together? | 1 |
| (ii) In the order A, E, I, O, U, throughout the arrangement?
(not necessarily next to each other) | 1 |
| <i>Answers may be left unsimplified.</i> | |
|
 | |
| (c) If $\sin A = \frac{2}{3}$ and $90^\circ < A < 180^\circ$, find the exact value of $\tan 2A$. | 3 |
|
 | |
| (d) A parabola has the equation $8y = x^2 + 6x + 1$. | |
| (i) Write the equation of the parabola in the form:
$(x - h)^2 = 4a(y - k)$. | 2 |
| (ii) Draw a neat sketch of the parabola on a number plane showing the: | |
| (α) Vertex | 1 |
| (β) Focus | 1 |
| (γ) Directrix | 1 |
|
 | |
| (e) Write $\cos \theta + \sqrt{3} \sin \theta$ in the form $R \cos(\theta + \alpha)$,
where $0^\circ < \alpha < 360^\circ$ and $R > 0$. | 2 |

QUESTION 2: START A NEW PAGE

- (a) Use the substitution $t = \tan\left[\frac{\theta}{2}\right]$ to solve to the nearest minute:

Marks
4

$$2 \sin \theta - \cos \theta = 1 \quad \text{for } 0^\circ \leq \theta \leq 360^\circ.$$

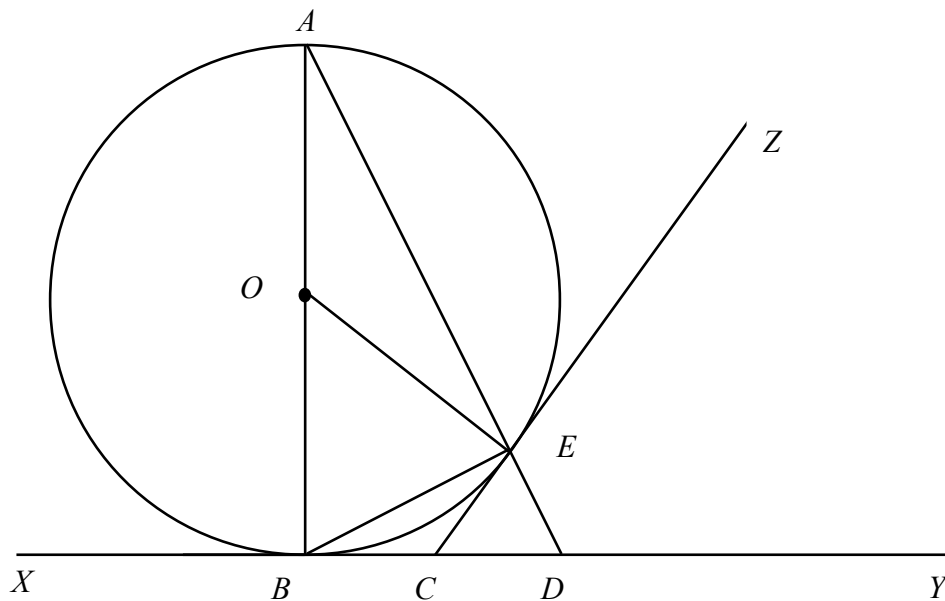
- (b) (i) Sketch the graph of $y = \frac{x-2}{x-4}$.

2

- (ii) Hence or otherwise, solve $\frac{x-2}{x-4} \leq 2$.

2

- (c) XY and CZ are tangents at B and E respectively to circle centre O . Chord AE is produced to meet XY at D . E is joined to centre O and B .



- (i) Copy the diagram and prove that $\angle EOB = \angle ECD$.

3

- (ii) Show that C is the centre of the circle through points B , E and D .

4

QUESTION 3: START A NEW PAGE

Marks

- (a) If α , β and γ are the roots of the equation $2x^3 - 3x^2 - 4x + 1 = 0$ find the value of:

(i) $\alpha + \beta + \gamma$

1

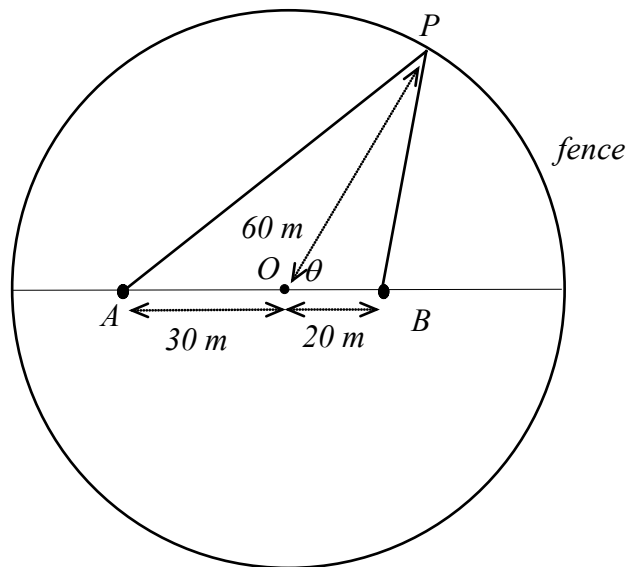
(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

2

- (b) Find the general solution in degrees to $\sin 2\theta - \cos \theta = 0$.

3

- (c) In an activity during a sport class, students are to run from position A to the fence surrounding a circular field of radius 60 m and return to position B . Jason, who is a top Maths student, decides to calculate which direction will give him the shortest path.



- (i) If he runs from A to P then to B show that the total path length D m is given by:

2

$$D = 30\sqrt{5 + 4\cos\theta} + 20\sqrt{10 - 6\cos\theta},$$

where $\angle POB = \theta$ and $0 \leq \theta \leq \pi$.

- (ii) Find the value of θ which will give Jason the shortest path. Justify your answer.

5

- (d) 16 people are invited to a party. They are to sit around 2 circular tables, with 8 people at each table.

- (i) How many different ways can 2 groups of 8 be chosen from the 24 people?

1

- (ii) Find the number of different seating arrangements that are possible at the party?

1

Answers may be left unsimplified.

QUESTION 4: START A NEW PAGE

Marks

- (a) When the polynomial $P(x)$ is divided by $x - 1$, the remainder is 2, and when divided by $x - 2$ the remainder is 1. What is the remainder when $P(x)$ is divided by $(x - 1)(x - 2)$?

3

- (b) Find the Cartesian equation of the curve whose parametric equations are:

3

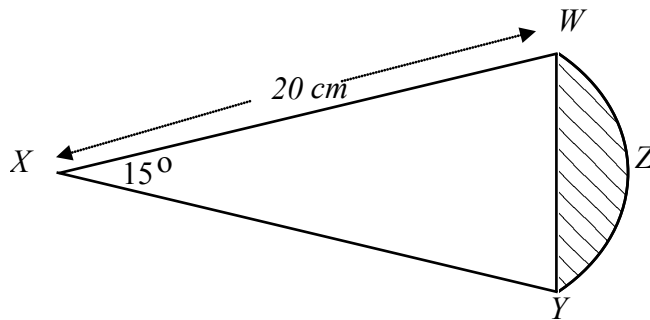
$$x = \cos 2t \text{ and } y = \cos t, \quad 0 \leq t \leq 2\pi.$$

- (c) (i) Show that the exact value of $\sin 15^\circ$ is $\frac{\sqrt{3} - 1}{2\sqrt{2}}$.

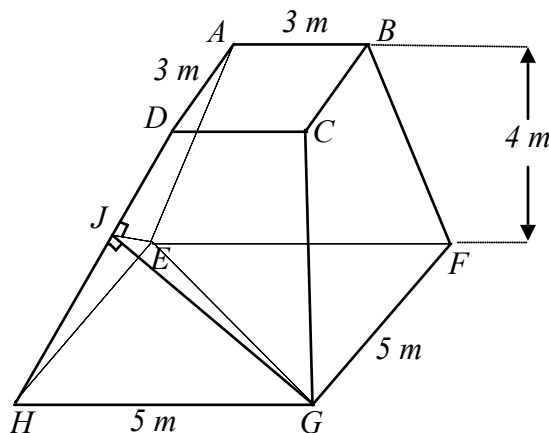
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- (ii) If $WXYZ$ is a sector of a circle of radius 20 cm, find the exact area of the segment shaded in the diagram below.

2



- (d) A frustrum of a right square pyramid has a base of side length 5 m, and a square top of side length 3 m. Its vertical height is 4 m.



- (i) Show that the length of the slant edge DH is $3\sqrt{2}$ m.

2

- (ii) Calculate the size of $\angle GJE$, the angle between the sloping faces, to the nearest minute.

3

END OF EXAMINATION

3 (a) x, β, γ roots of $2x^3 - 3x^2 - 4x + 1 = 0$

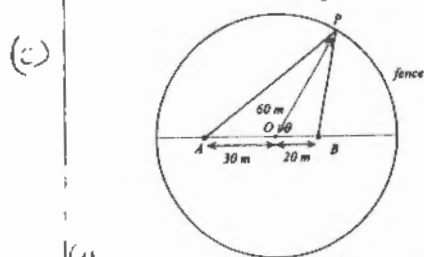
(i) $x + \beta + \gamma = -\frac{-b}{a} = \frac{3}{2}$ ①

(ii) $x\beta + \beta\gamma + \gamma x = \frac{c}{a} = -\frac{4}{2} = -2$

$x\beta\gamma = \frac{d}{a} = -\frac{1}{2}$

$\frac{1}{x} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \gamma x + x\beta}{x\beta\gamma} = \frac{-2}{-\frac{1}{2}} = 4$ ①

(b) $\sin 2\theta - \cos \theta = 0$
 $2\sin \theta \cos \theta - \cos \theta = 0$
 $\cos \theta (2\sin \theta - 1) = 0$ ①
 $\cos \theta = 0$ or $\sin \theta = \frac{1}{2}$
 $\theta = 360^\circ \pm 90^\circ$ or ①
 $\theta = 180^\circ + (-1)^n 30^\circ$
 n is an integer ①



(c) $AP^2 = 30^2 + 60^2 - 2 \times 30 \times 60 \cos(150^\circ)$
 $= 900 + 3600 + 3600 \cos \theta$
 $= 4500 + 3600 \cos \theta$
 $AP = \sqrt{4500(5 + 4 \cos \theta)}$ ①
 $AP^2 = 20^2 + 60^2 - 2 \times 20 \times 60 \cos \theta$
 $= 400 + 3600 - 2400 \cos \theta$
 $PB = \sqrt{400(10 - 6 \cos \theta)}$ ①
 $= 20 \sqrt{10 - 6 \cos \theta}$ ①

$\therefore D = AP + PB$
 $= 30 \sqrt{5 + 4 \cos \theta} + 20 \sqrt{10 - 6 \cos \theta}$
 $\therefore D = 30(5 + 4 \cos \theta)^{\frac{1}{2}} + 20(10 - 6 \cos \theta)^{\frac{1}{2}}$
 $\frac{dD}{d\theta} = 30 \times \frac{1}{2} (5 + 4 \cos \theta)^{-\frac{1}{2}} (-4 \sin \theta)$
 $+ 20 \times \frac{1}{2} (10 - 6 \cos \theta)^{-\frac{1}{2}} (6 \sin \theta)$
 $= \frac{-60 \sin \theta}{\sqrt{5 + 4 \cos \theta}} + \frac{60 \sin \theta}{\sqrt{10 - 6 \cos \theta}}$
 $= -60 \sin \theta \left[\frac{1}{\sqrt{5 + 4 \cos \theta}} - \frac{1}{\sqrt{10 - 6 \cos \theta}} \right]$
 for stationary point $\frac{dD}{d\theta} = 0$
 $\therefore \sin \theta = 0$ or $\frac{1}{\sqrt{5 + 4 \cos \theta}} = \frac{1}{\sqrt{10 - 6 \cos \theta}}$
 $\sin \theta = 0 \Rightarrow \theta = 0$ or π (endpoints) ①
 $\sqrt{5 + 4 \cos \theta} = \sqrt{10 - 6 \cos \theta}$
 $5 + 4 \cos \theta = 10 - 6 \cos \theta$
 $10 \cos \theta = 5$
 $\cos \theta = \frac{1}{2}$
 $\therefore \theta = \frac{\pi}{3}$

Test stationary point at $\frac{\pi}{3}$
 (or calculate distance for $\theta = \frac{\pi}{3}$)

θ	$\frac{\pi}{3}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\frac{dD}{d\theta}$	3.56	0	-7.9
	2.72	0	-

function is continuous for $0 < \theta < \pi$, gradient changes from positive to negative and is stationary point at $\theta = \frac{\pi}{3}$
 $\therefore$ local MAX at $\theta = \frac{\pi}{3}$

$\therefore$ As there is only one stationary point $0 < \theta < \pi$ absolute min must be at one endpoint i.e. $\theta = \pi$ or $\theta = 0$
 $\theta = 0 \quad D = 30 \sqrt{5+4} + 20 \sqrt{10-6}$
 $= 130 \text{ m}$
 $\theta = \pi \quad D = 30 \sqrt{5-4} + 20 \sqrt{10+6}$
 $= 110 \text{ m}$
 $\therefore$ Min distance is when $\theta = \pi$ (Alternative to testing $\frac{dD}{d\theta}$ at $\theta = \frac{\pi}{3}$ - calculate the dist = 132.3 m) $\therefore$ longer not min

3 (d) 2 groups of 8 from 16 people (groups both equal)
 $\frac{{}^{16}C_8 \times {}^8C_8}{2!} = \frac{12870}{2}$
 $= 6435$ ①

Number of seating arrangements at each table is 7!

$\therefore$ Total no. at 2 tables is

$\frac{{}^{16}C_8 \times {}^8C_8}{2!} \times (7!)^2$ ①

4 (a) $P(x) = (x-1)(x-2)Q(x) + ax + b$

$P(1) = 2$ ①
 $P(2) = 1$ ①
 $2 = a + b$
 $1 = 2a + b$ ①

$\therefore 1 = -a$
 $a = -1$
 $b = 3$

$\therefore$ Remainder is $-x + 3$ ①

(b) $x = \cos 2t$
 $-1 \leq x \leq 1$

$y = \cos t$
 $-1 \leq y \leq 1$

$x = 2y^2 - 1$ ①

$x = 2y^2 - 1$ ①

or $2y^2 = x + 1$ ①

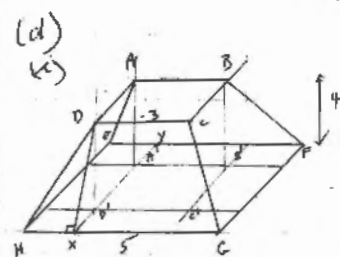
$-1 \leq x \leq 1$ ①

or $(-1 \leq y \leq 1)$

4 (c) $\sin(45^\circ - 30^\circ) = \sin 15^\circ$
 $= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$ ①
 $= \frac{\sqrt{3} - 1}{2\sqrt{2}}$
 $= \frac{\sqrt{3} - 1}{2\sqrt{2}}$

Area of segment

$A = \frac{1}{2} r^2 (\theta - \sin \theta)$
 $= \frac{1}{2} \times 20^2 \left(\frac{15 \times \pi}{180} - \sin 15^\circ \right)$
 $= 200 \left(\frac{\pi}{12} - \left(\frac{\sqrt{3} - 1}{2\sqrt{2}} \right) \right)$ ①
 cm^2

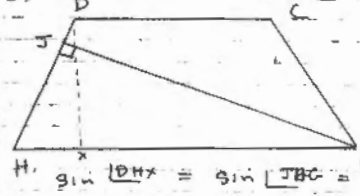


$XD' = A'Y = HX$
 $= \frac{5-3}{2}$
 $= 1 \text{ cm}$

$\therefore XD' = \sqrt{4^2 + 1^2}$
 $= \sqrt{17} \text{ cm}$ ①

$HX = 1 \text{ cm}$
 $DH = \sqrt{17+1}$
 $= \sqrt{18} \text{ cm}$

$\therefore DH = \frac{3\sqrt{2}}{2} \text{ cm}$ ①
 $\therefore \sin \angle DHX = \frac{1}{\sqrt{2}}$



$\therefore \sin \angle DHX = \sin \angle BGC = \frac{1}{\sqrt{2}}$

Extension I 2005

1. (a) $\sqrt{3}x + y = 2$ $m = -\sqrt{3}$ ①

$x - y = 0$ $m = 1$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{-\sqrt{3} - 1}{1 - (\sqrt{3})(1)} \right|$ ①

$\theta = 75^\circ$ ①

(b) (i) AEIOU ② ③ ④ ⑤ ⑥

6 elements arranged 6! ways

A, E, I, O, U can be arranged

5! ways

$\therefore$ total arrangements

$= 6! \times 5! = 86400$

ways ①

(ii) 10 elements arranged 10! ways

For each arrangement of vowels only one has them in the correct order i.e. 1 out of 5!

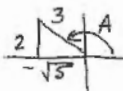
$\therefore$ no of ways with vowels in AEIOU order = 10!

$= \frac{10!}{5!} = 30240$ ①

(c) $\sin A = \frac{2}{3}$ $90^\circ < A < 180^\circ$

$\therefore$ 2nd quadrant angle.

$\therefore \tan A < 0$ $\tan A = -\frac{2}{\sqrt{5}}$ ①



$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

① $= \frac{2(-\frac{2}{\sqrt{5}})}{1 - (\frac{4}{5})} = -\frac{4}{\sqrt{5}}$

① $= -4\sqrt{5}$

(d)

$8y = x^2 + 6x + 1$

$\therefore x^2 + 6x = 8y - 1$

$x^2 + 6x + 9 = 8y - 1 + 9$

$(x+3)^2 = 8(y+1)$

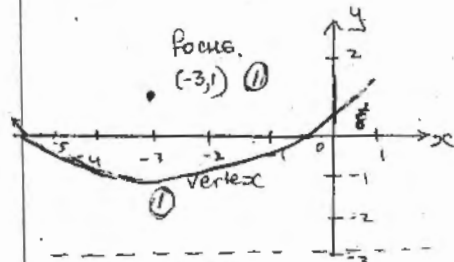
$(x+3)^2 = 4 \times 2(y+1)$ ①

vertex = $(-3, -1)$

$\therefore$ focal length = 2 units

$\therefore$ focus = $(-3, 1)$

directrix $y = -3$



① $y = -3$ directrix

(e)

$\cos \theta + \sqrt{3} \sin \theta$

$= 2 \left[\frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta \right]$

$= R [\cos \theta \cos \alpha - \sin \theta \sin \alpha]$

$R^2 = 1^2 + (\sqrt{3})^2 = 4$

$\therefore R = 2$ (as $R > 0$)

$\cos \alpha = \frac{1}{2}$ $\sin \alpha = -\frac{\sqrt{3}}{2}$

$\therefore \alpha$ is a 4th quad angle

$\therefore \alpha = 360^\circ - 60^\circ$

$= 300^\circ$

$\therefore$ Answer is

$2 \cos(\theta + 300^\circ)$

① ①

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2

(a) $t = \tan(\frac{1}{2}\theta)$

$\therefore \sin \theta = \frac{2t}{1+t^2}$ $\cos \theta = \frac{1-t^2}{1+t^2}$

$\therefore 2 \sin \theta - \cos \theta = 1$ ①

$\therefore 2 \left(\frac{2t}{1+t^2} \right) - \left(\frac{1-t^2}{1+t^2} \right) = 1$

$\therefore 4t - 1 + t^2 = 1 + t^2$

$4t = 2$

$t = \frac{1}{2}$

$\tan \frac{1}{2}\theta = \frac{1}{2}$ ①

$\therefore \frac{1}{2}\theta = 26.5^\circ 55' 11.8''$

$\therefore \theta = 53.13010236^\circ$

$\theta = 53^\circ 8'$ (nearest minute)

check $180^\circ = \theta$

LHS $= 2 \times \sin 180^\circ - \cos 180^\circ$

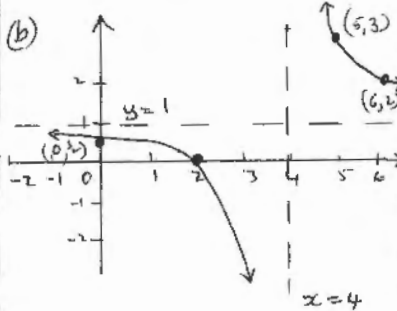
$= -(-1) = 1$

$=$ RHS

$\therefore 180^\circ$ is a solution

$\therefore \theta = 53^\circ 8' \text{ or } 180^\circ$

① ①



asymptotes ①

intercepts ①

to solve $\frac{x-2}{x-4} \leq 2$

consider $\frac{x-2}{x-4} = 2$

$x - 2 = 2x - 8$

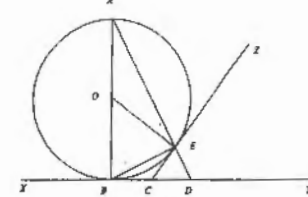
$6 = x$

From graph

$x < 4$ or $x \geq 6$ ①

①

(c)



(i) To prove $\angle EOB = \angle ECD$

① $\angle OBC = 90^\circ$ (angle between tangent and radius at B)

$\angle OEC = 90^\circ$ (similarly at E)

① $\therefore OECD$ is a cyclic quadrilateral (opposite angles of $OECD$ sum to 180°)

① $\therefore \angle ECB = \angle EDC$ (exterior angle of $OECD$ equals opposite interior angle)

(ii) To show C is centre of circle through B, E and D.

① $CE = CB$ (intercepts of tangents from external point C are equal)

$\angle AEB = 90^\circ$ (angle subtended by diameter AB at circumference is 90°)

$\angle CED + \angle BEC + \angle AEB = 180^\circ$ (angle sum of straight angle at E is 180°)

$\therefore \angle CED + \angle BEC = 90^\circ$

$\angle BEC = \angle DAB$ (angle in alternate segment is equal to angle between tangent and chord at E)

① $\therefore \angle CED + \angle DAB = 90^\circ$

$\angle ABD = 90^\circ$ (angle between radius and tangent = 90° at B)

$\angle ABD + \angle DAB + \angle ADB = 180^\circ$ (angle sum of $\triangle ABD$ is 180°)

① $\therefore \angle DAB + \angle ADB = 90^\circ$

$\therefore \angle CED = \angle ADB$

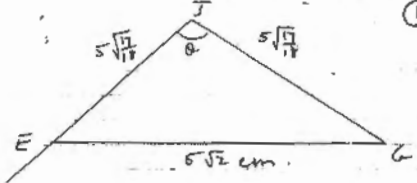
① $\therefore CE = CD$ (equal sides angles in $\triangle CED$)

$\therefore CE = CD = CB$

$\therefore$ C is centre of circle through B, E, D.

$$\therefore JG = HG \times \sin \angle H$$

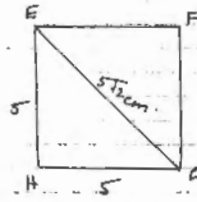
$$= 5 \times \sqrt{\frac{17}{18}} \quad (\text{can also do with similar triangle})$$



By symmetry $JE = JG$.

$$EG = \sqrt{5^2 + 5^2}$$

$$EG = 5\sqrt{2} \text{ cm}$$



$$\cos \angle JGE = \frac{(5\sqrt{\frac{17}{18}})^2 + (5\sqrt{\frac{17}{18}})^2 - (5\sqrt{2})^2}{2 \times 5\sqrt{\frac{17}{18}} \times 5\sqrt{\frac{17}{18}}} \quad (1)$$

$$= \frac{25 \times \frac{17}{18} + 25 \times \frac{17}{18} - 25 \times 2}{2 \times 25 \times \frac{17}{18}}$$

$$= \frac{25 \left(\frac{17}{18} + \frac{17}{18} - 2 \right)}{25 \times \frac{17}{9}}$$

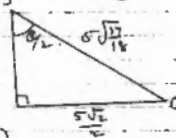
$$= \frac{17 - 2}{9}$$

$$= \frac{17 - 18}{17}$$

$$= -\frac{1}{17}$$

$$\therefore \angle JGE = 93.22^\circ \quad (1)$$

Alternatively



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